

4.6 Descriptive Statistics

اوسط سب کا مجموعہ تقسیم تعداد، میڈین ترتیب کے بعد بیچ، موڈ سب سے زیادہ بار آنے والا۔

CS XII

Chapter 4: Data and Analysis

Lecture 08 of 8

★ Golden

4.6 Descriptive Statistics ★ Golden

LEARN IT

Descriptive statistics squeeze a long list into a few honest numbers. **Central tendency** asks “what is typical?”. **Spread** asks “how mixed is the list?”.

4.6.1 Mean, median, mode

Measure	How	Use when
Mean	Sum ÷ n. Uses every value.	Values sit close together. Example: ordinary test scores.
Median	Middle after sorting. Odd n: centre. Even n: mean of the two centre values.	Outliers would wreck the mean (CEO salary).
Mode	Most frequent value. May be none, one, or several.	Most popular choice (shoe size).

WORKED EXAMPLE

Marks 70, 80, 90. Mean = $240/3 = 80$.

Odd list 35, 10, 15, 70, 95. Sort: 10, 15, 35, 70, 95. Median = **35**.

Bottles sold 20, 25, 20, 30, 20, 25, 20. Mode = **20** (appears four times).

Outlier trap: nine people earn 10, one earns 1000. Mean = $1090/10 = 109$ (looks rich). Median = 10 (tells the truth).

4.6.2 Spread: range, variance, standard deviation, min, max

- **Min / max** — smallest and largest. Marks 70, 80, 90, 60, 75 → min 60, max 90.
- **Range** = max – min. Sales 120, 150, 180 → range 60.
- **Population variance** $\sigma^2 = \frac{1}{n} \sum (x_i - \mu)^2$.
- **Sample variance** s^2 uses $\div (n - 1)$. `pandas.Series.std()` uses the sample version.
- **Standard deviation** $\sigma = \sqrt{\sigma^2}$. Same units as the data (marks, not marks²). Low SD → tight, consistent. High SD → spread out, unpredictable.

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2, \quad \sigma = \sqrt{\sigma^2}, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

WORKED EXAMPLE

Full working on 70, 80, 90. Mean $\mu = 80$. Deviations -10, 0, +10. Squares 100, 0, 100. Sum = 200.

Population variance $\sigma^2 = 200/3 = \mathbf{66.67}$ (2 d.p.). Population SD $\sigma = \sqrt{200/3} = \mathbf{8.16}$ (2 d.p.).

Sample variance $s^2 = 200/2 = \mathbf{100}$. Sample SD $s = 10$. That is what `pd.Series([70,80,90]).std()` returns.

EXAM TIP

Always state whether you used n or $n-1$. Show the sorted list for the median. For SD, show: mean $\rightarrow$ deviations $\rightarrow$ squares $\rightarrow$ divide $\rightarrow$ square root. A box plot pictures min, Q1, median, Q3, max of the same data.

COMMON MISTAKE

Do not report variance as if it were in "marks". Variance is marks^2 . Quote the standard deviation when you want a sentence in the original units.

اردو میں سمجھیں

اوسط سب کا مجموعہ تقسیم تعداد، میڈین ترتیب کے بعد بیچ، موڈ سب سے زیادہ بار آنے والا۔ رینج بڑا منفی چھوٹا۔ ویرینس اوسط سے فاصلوں کے مربع کی اوسط ہے، معیاری انحراف اس کا جذر ہے اور اصل اکائی میں ہوتا ہے۔ پینڈاز `std()` (نمونے والا $n-1$) استعمال کرتا ہے۔

CHECK YOURSELF

1. Median of 4, 8, 6, 10, 2?
2. Range of 120, 150, 180?
3. Does pandas `.std()` divide by n or $n-1$?

Answers: 1. Sorted 2,4,6,8,10 $\rightarrow$ 6. 2. 60. 3. $n-1$ (sample).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Mean of 70,80,90 is:

- a. 80
- b. 70
- c. 90
- d. 240

MCQ 2. Median of 10,15,35,70,95 is:

- a. 35
- b. 10
- c. 70
- d. 45

MCQ 3. Mode of 20,25,20,30,20,25,20 is:

- a. 20
- b. 25
- c. 30
- d. 7

MCQ 4. Range of 120,150,180 is:

- a. 60
- b. 30
- c. 150
- d. 180

MCQ 5. pandas .std() uses:

- a. $n-1$ (sample)
- b. n always
- c. $2n$
- d. zero

MCQ 6. SD is the square root of:

- a. Variance
- b. Mean
- c. Median
- d. n

Extra short questions

1. When mean vs median vs mode.
2. Population variance formula idea.
3. Why square root for SD?
4. Outlier trap with 9×10 and one 1000.

Extra long question

1. For 4,8,6,10,2 compute mean, median, range, population variance and SD. Then say what pandas std reports. Show every step. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Mean typical no outliers; median with outliers; mode popular category/value.
2. Mean of squared deviations; $\div n$. Sample $\div (n-1)$.
3. Back to original units.
4. Mean 109 vs median 10.

Long

1. Sorted 2,4,6,8,10. Mean 6, median 6, range 8. Devs -4, -2, 0, 2, 4 sq 16, 4, 0, 4, 16 sum 40. $\sigma^2=8$, $\sigma=\sqrt{8}\approx 2.828$. pandas $40/4=10$, $\sqrt{10}\approx 3.162$.